IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KANNAN AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

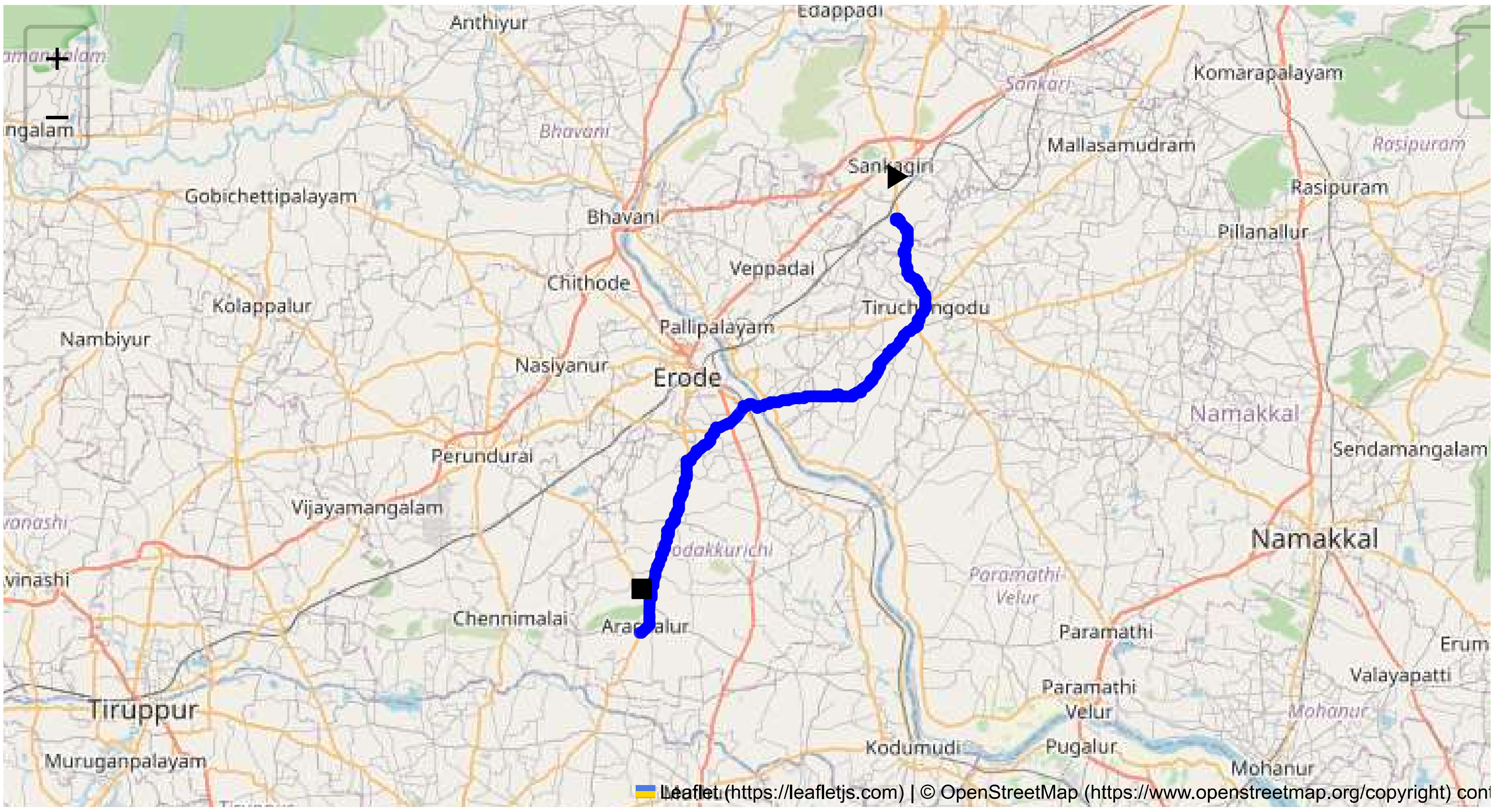
Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 46.34 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.153792, 77.694313)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

- The route from Sangagiri to Erode spans approximately 46.34 kilometers, traversing a combination of state highways and local roads in Tamil Nadu. The journey typically takes around 1 hour and 18 minutes for heavy vehicles due to traffic and road conditions.

2. Typical Weather Conditions

- **Weather Patterns**: Tamil Nadu generally experiences tropical weather with hot summers, moderate to heavy monsoons, and mild winters.
- **Weather Hazards**:
  - **Monsoon**: Heavy rainfall can lead to waterlogging and reduced visibility, impacting driving conditions.
  - **Summer**: High temperatures may result in overheating of vehicle engines and discomfort for drivers.
- It is essential to monitor weather forecasts regularly and prepare for sudden changes, especially during the monsoon season.

3. Traffic Patterns and Congestion

- **Peak Hours**: Typical traffic congestion occurs during the early morning (7-9 AM) and late afternoon to evening (5-8 PM).
- **Congestion Zones**: Urban areas, especially near town intersections and market locations, may experience slow-moving traffic. Erode sees higher congestion due to its commercial nature.

4. Road Quality and Infrastructure

- **Road Conditions**: While most state highways maintain a decent quality, local roads may have potholes and uneven surfaces. Watch for construction zones or repairs.
- **Infrastructure**: Bridges, narrow lanes, and urban bottlenecks could present challenges. Ensure vehicle compliance with bridge weight limits and lane widths.

5. Alternative Routes for Emergencies

- **Emergency Route**: Via NH544 - A major highway providing a more expeditious connection between the two locations. It offers better quality infrastructure and more predictable driving conditions in emergencies.

6. Local Regulations on Hazardous Material Transport

- **Regulations**: Transporters must adhere to state regulations regarding signage, material safety data sheets, and emergency response information. Certain routes may be restricted during specific hours for hazardous material transit.

7. Historical Incidents

- **Incident Records**: Historical data indicates a few accidents involving heavy vehicles in the region, typically due to over-speeding or overloading. Past incidents necessitate cautious driving and adherence to traffic rules.

8. Environmental Considerations

- **Sensitive Areas**: The route passes through agricultural areas where pollution control is critical. Avoid spills or emissions.
- **Conservation Areas**: Be conscious of wildlife crossings and preserve local fauna by reducing speed near designated zones.

9. Communication Coverage

- **Coverage Assessment**: The route generally has good mobile connectivity, but some rural stretches may have weak signals. It is advisable to plan for such areas by ensuring alternative communication methods, like radios, are functional.

10. Emergency Response Times

- **Response Efficiency**: Urban areas like Erode have quicker emergency services. However, rural segments of the route could see slower response times; keep local emergency numbers handy.

11. Overall Risk Assessment

- **Risk Level**: Moderate
- **Key Risks**: Weather-related hazards, congestion, and road quality issues can pose challenges. Proper pre-trip planning and real-time adjustments based on condition updates will mitigate these risks.

In conclusion, while the route from Sangagiri to Erode is generally manageable for heavy vehicles transporting hazardous materials, attention to weather, traffic, and road quality, along with compliance with local regulations, will ensure a safe journey. Proactive emergency planning and communication will further enhance transportation safety on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	Medium	11.37842, 77.89452	30 KM/Hr
6	Turn	High	11.36604, 77.88887	15 KM/Hr
7	Turn	Medium	11.34067, 77.86347	30 KM/Hr
8	Turn	Medium	11.33882, 77.86218	30 KM/Hr
9	Turn	Medium	11.33778, 77.86236	30 KM/Hr
10	Turn	High	11.29490, 77.74685	15 KM/Hr
11	Turn	Medium	11.29141, 77.74662	30 KM/Hr
12	Turn	High	11.29129, 77.74657	15 KM/Hr
13	Turn	Medium	11.28844, 77.74319	30 KM/Hr
14	Turn	Medium	11.28573, 77.74281	30 KM/Hr
15	Turn	Medium	11.27336, 77.72999	30 KM/Hr
16	Turn	Medium	11.27319, 77.72981	30 KM/Hr
17	Turn	High	11.27328, 77.72705	15 KM/Hr
18	Turn	Medium	11.23054, 77.71796	30 KM/Hr
19	Turn	Medium	11.22387, 77.71385	30 KM/Hr
20	Turn	Medium	11.16078, 77.70122	30 KM/Hr
21	Turn	High	11.15942, 77.70068	15 KM/Hr
22	Turn	Medium	11.15935, 77.70060	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
10	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
13	hospital	Government Hospital	11.2343326, 77.719226	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37842, 77.89452

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36604, 77.88887

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.34067, 77.86347

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33882, 77.86218

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33778, 77.86236

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.29490, 77.74685

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.29141, 77.74662

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.29129, 77.74657

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.28844, 77.74319

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.28573, 77.74281

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.27336, 77.72999

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.27319, 77.72981

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.27328, 77.72705

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.23054, 77.71796

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.22387, 77.71385

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16078, 77.70122

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15942, 77.70068

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15935, 77.70060